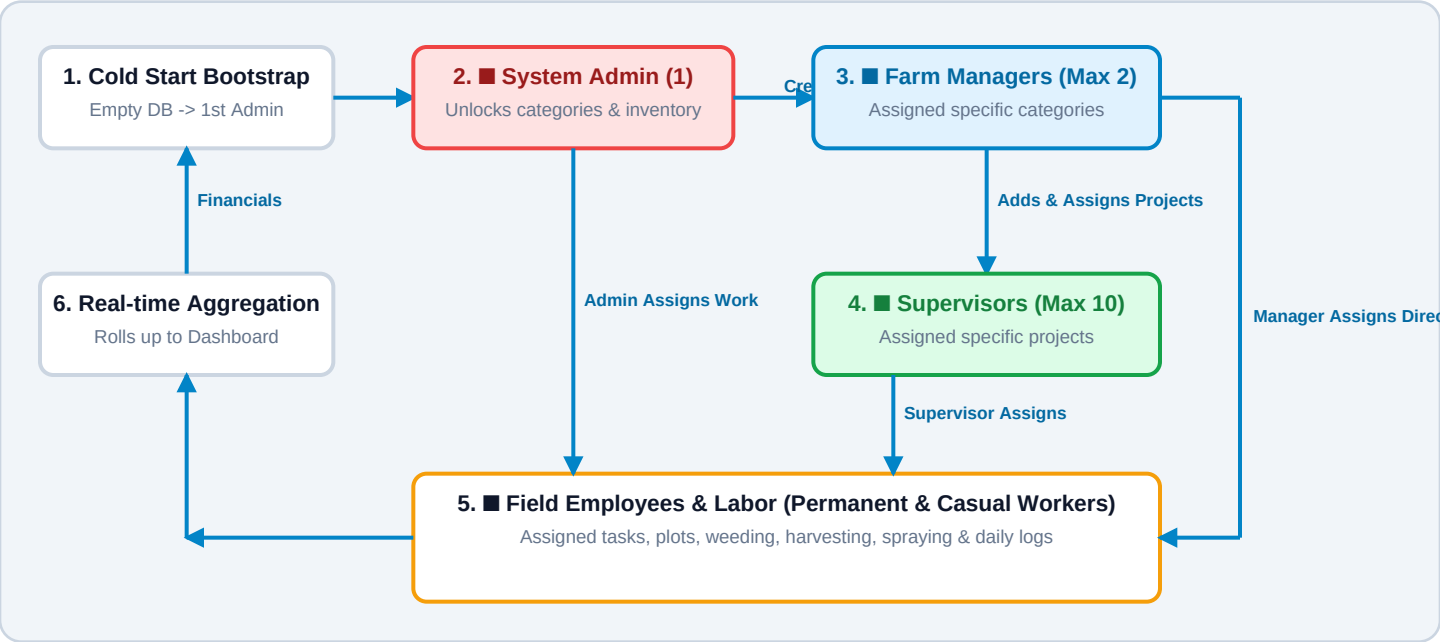


Access Control & Role Hierarchy Specification

Multi-Tier Scoped Permissions Flowchart · Admin, Manager, Supervisor & Employee Delegation

■ Admin (Superuser) Cap: 1 Account Scope: Entire Farm Platform Key Power: First bootstrap, user approval, system settings, can assign tasks to employees.	■ Manager Cap: Max 2 Accounts Scope: Assigned Categories/Projects Key Power: Adds supervisors, creates projects in assigned categories, assigns employee work.	■ Supervisor Cap: Max 10 Accounts Scope: Assigned Projects Only Key Power: Field operations, supplies usage, harvests, manages and assigns work to workers.	■ Employees / Labor Cap: Unlimited Type: Casual / Permanent Assigned By: Admin, Manager, or Supervisor for field tasks and operations.
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System Lifecycle & Access Delegation Flowchart



Detailed Role Permissions & Action Scope

Domain Area	■ Admin (1)	■ Manager (Max 2)	■ Supervisor (Max 10)	■ Laborers
Initial Cold Start / Bootstrap	Full Setup (First Account)	Blocked until Admin exists	Blocked until Manager invites	N/A
Categories Management	Create, Edit, Delete, Assign	View Assigned Categories	No category access	N/A
Projects Lifecycle	All projects across farm	Create & manage in assigned cat.	View & run assigned projects	Execute tasks
Worker / Task Assignment	Can assign any employee	Can assign branch employees	Can assign project employees	Assigned to

Supplies & Inventory	Restock warehouse, pricing	Restock warehouse, pricing	Use supplies on project only	Uses tools
Harvest & Sales	All records & invoicing	Assigned categories sales	Record Harvest & usage logs	Field labor
Financial & Metrics Dashboard	Farm-wide budget & profits	Scoped to assigned domain	Scoped to assigned project	No access

■ Core Backend Security & Quota Invariants

- **Hard Quotas:** Spring Boot strictly enforces `countByRole('ADMIN') <= 1`, `countByRole('MANAGER') <= 2`, `countByRole('SUPERVISOR') <= 10`.
- **Tri-Tier Labor Assignment:** Admin, Manager, and Supervisor can all assign daily or permanent tasks to field workers.
- **Database-Level Query Scoping:** All queries automatically filter by manager/supervisor foreign keys (tamper-proof URL security).